

JAVA PRACTICAL RECORD

Experiment No.	03
Date	29/08/2026
Program	Farm Animals
Concept	Classes, Objects, Constructor and Methods

1. Objective / Aim

To design and implement a Java program using a class, constructor, objects, and a method to store and display the details of different farm animals.

2. System Requirements

Hardware: Computer/Laptop with basic processing capability.

Software: Java Development Kit (JDK) 8 or above and any Java IDE or command-line terminal.

3. Algorithm / Logic Description

1. Start the program.
2. Create a class **FarmAnimal** with fields for name, place of stay, food, and sound.
3. Define a constructor to initialize all four fields.
4. Define the **displayDetails()** method to print the animal information.
5. Create three objects: **cow**, **pig**, and **horse** using the constructor.
6. Call **displayDetails()** for each object.
7. Display the details of all three farm animals.
8. Stop the program.

4. UML Class Diagram

FarmAnimal	FarmAnimals
- name : String - stays : String - eats : String - sound : String	+ main(args : String[]) : void
+ FarmAnimal(name : String, stays : String, eats : String, sound : String) : FarmAnimal + displayDetails() : void	Creates FarmAnimal objects: cow, pig, horse Calls displayDetails()

Relationship: **FarmAnimals** is the driver class. Its main() method creates and uses multiple objects of **FarmAnimal**.

5. Source Code

```
class FarmAnimal {
    String name;
    String stays;
    String eats;
    String sound;

    // Constructor
    FarmAnimal(String name, String stays, String eats, String sound) {
        this.name = name;
        this.stays = stays;
        this.eats = eats;
        this.sound = sound;
    }

    // Method to display animal details
    void displayDetails() {
        System.out.println("Animal : " + name);
        System.out.println("Stays : " + stays);
        System.out.println("Eats : " + eats);
        System.out.println("Sound : " + sound);
        System.out.println("-----");
    }
}

public class FarmAnimals {
    public static void main(String[] args) {

        // Creating objects
        FarmAnimal cow = new FarmAnimal(
            "Cow", "Cowshed", "Grass and Hay", "Moo"
        );

        FarmAnimal pig = new FarmAnimal(
            "Pig", "Pigsty", "Grains and Vegetables", "Oink"
        );

        FarmAnimal horse = new FarmAnimal(
            "Horse", "Stable", "Grass and Hay", "Neigh"
        );

        // Displaying details
        cow.displayDetails();
        pig.displayDetails();
        horse.displayDetails();
    }
}
```

6. Compilation & Execution Commands

```
javac FarmAnimals.java
```

```
java FarmAnimals
```

7. Output Screen

```
Animal : Cow
Stays : Cowshed
Eats : Grass and Hay
Sound : Moo
-----
Animal : Pig
Stays : Pigsty
Eats : Grains and Vegetables
Sound : Oink
-----
Animal : Horse
Stays : Stable
Eats : Grass and Hay
Sound : Neigh
-----
```

8. Result

The Java program was successfully designed and executed using a class, constructor, objects, and a method. The details of the cow, pig, and horse were displayed successfully.